

Indo-Swiss Research Database - Data import and processing log

This R Markdown Notebook is running and logging the import of data from raw CSV files and carrying out initial processing. It makes note of all analysis and cleaning decisions made while gathering the data.

Summaries of the initial data files

```
## The Web of Science data consists of 15806 entries, ranging from 2000 to 2024.
```

```
## The Scopus data consists of 17229 entries, ranging from 2000 to 2025.
```

We are now exploring how to match the datasets. One easy way to match is to use doi (digital object identifiers), which are created for this express purpose

```
## For the WoS data, 13969 out of 15806 entries have a doi. That corresponds to 88%.
```

```
## Within this dataset, there are 38 dois that are not unique. Looking in and cleaning them.
```

```
## Mostly these are situations where there are small variations and duplications. We clean these duplications.
```

```
## These are cleaned now, and we have 13931 publications with a unique doi from Web of Science.
```

```
## For the Scopus data, 16493 out of 17229 entries have a doi. That corresponds to 96%.
```

```
## Within this dataset, there are 27 dois that are not unique. Looking into them deeper.
```

```
## Mostly these are situations where the same authors have written the preface to a book and a chapter within it.
```

```
## These are cleaned now, and we have 16466 publications with a unique doi from Scopus.
```

Now let us try to match the two databases based on doi

```
## When we match entries between the two databases using DOI, we find
```

```
## 11760 entries common to both databases
```

```
## 4706 entries only in Scopus
```

```
## 2171 entries only in Web of Science.
```

```
##
```

```
## There are 9 matched papers where there is a difference of publication date of more than one year between the two datasets.
```

```
##
```

```
## For the fields - article title, document type, publisher, isbn, issn, issue, open access, volume, publication year, we have matched the two datasets.
```

```
##
```

```
## Now this dataset of 18637 entries all with DOIs and matched between Scopus and Web of Science, is what we will use for further analysis.
```

```
#Processing Author Information
```

Both Scopus and Web of Science provide their own unique identifiers for authors. We need to process the author data from each database separately, and then combine them.

```
## Proceeding with only a subset of 500 publications, in order to check that the code is working. Once we are confident, we will process the full dataset.
```

```
##
```

First, processing the author data from Web of Science.

```
## Extracting data of individual authors from each publication. Using the library 'countries' to assign country codes to each author.
```

```
##
```

```
##
## In total 143 unique country names were provided
## 127/143 have been matched with EXACT matching
## 16/143 have been matched with FUZZY matching, out of which:
## 4/16 are a POOR match (likely wrongly identified)
##
##
## Multiple arguments have been matched to the same country name:
##   - Estonia : Estonia
##   - Fr : France
##   - France : France
##   - Por : Portugal
##   - Portugal : Portugal
##   - Scotland : Estonia
##   - Turkey : Turkey
##   - Turkiye : Turkey
##
## No close match found for the following countries, kept the original name:
## (set - poor_matches - to TRUE if you want the closest match to be returned or set - na_fill - to TRUE)
##   - CA USA; ;
##   - En
##   - OH 43210
##   - Peoples R China
```

Second, processing the author data from Scopus. Scopus also provides a unique numerical ID to each author. There are some challenges to uniquely extracting that (including the issue of there being multiple people named “M. Kumar” on the same paper, but with different Scopus IDs). So I will skip extracting that for now and add it in at a later stage.

```
dat1 = allentries %>% filter(datasource != 'WoS') %>% select(pubNum, `authors.s`, `author full names.s`, a
allinst = tibble()

printo("Extracting data of individual authors from each publication. Using the library 'countries' to a

## Extracting data of individual authors from each publication. Using the library 'countries' to assign
##
for (i in 1:nrow(dat1)) {
  # authors = dat1$`Author Full Names`[i] %>%
  #   str_split(pattern = ';', simplify=T) %>% str_trim()
  country_match = paste0("\\s*(", paste(list_countries(), collapse="|"), "|United States)")
  inst = dat1$addresses[i] %>% str_split(pattern = ';', simplify = T) %>% str_squish()
  inst = tibble(inst = inst) %>%
    mutate(author = str_extract(inst, '^[^,]+') %>% str_trim(),
           institution = str_extract(inst, '(?<=,).*') %>% str_trim()) %>%
    #There is a problem. Scopus includes multiple institutions in the field for the same author, with n
    mutate(institution = str_replace_all(institution, country_match, ", \\1<<<SPLIT>>>") %>% str_remove
    # mutate(singleauthor = author %>% str_split(pattern = '; '))
    separate_longer_delim(institution, '<<<SPLIT>>>') %>%
    mutate(institution = str_remove(institution, '^,') %>% str_trim())
  inst = inst %>%
    select(-inst) %>% mutate(pubNum = dat1$pubNum[i])
  allinst = bind_rows(allinst, inst)
  rm(inst)
}
```

```

# authors = dat1 %>% select(pubNum,author = authors.s) %>% separate_longer_delim(author,','; ')
# scopus_ids = dat1 %>% select(pubNum,aut = `author full names.s`) %>%
#   separate_longer_delim(aut,','; ') %>%
#   mutate(scopus_id = str_extract(aut,"(?<=\\()[0-9]+(?:=\\()")) %>% select(-aut)
# scopus_ids = cbind(authors,scopus_ids %>% select(-pubNum))# %>% distinct()
# rm(authors)

dat2 = dat1 %>%
  separate_longer_delim(`authors.s`,`','; ') %>% rename(author = authors.s) %>%
  # bind_cols(scopus_ids %>% select(scopus_id)) %>%
  select(-`author full names.s`, -addresses) %>%
  #ensure each author name comes only once
  distinct() %>%
  # left_join(scopus_ids,by=c('pubNum','author'),relationship = 'one-to-one') %>%
  left_join(allinst,by = join_by(pubNum, author), relationship = 'one-to-many') %>%
  ## PROBLEM - the hugely coauthored papers are having multiple authors who have the same initials, but
  mutate(country = str_extract(institution, '[^,]*$') %>% str_remove('^, ') %>%
    #str_remove('\\(data truncated to fit\\)') %>% str_remove('.*(?=USA$)') %>%
    country_name(to = 'simple',verbose = T, na_fill = T))

##
## In total 167 unique country names were provided
## 160/167 have been matched with EXACT matching
## 7/167 have been matched with FUZZY matching, out of which:
## 1/7 are a POOR match (likely wrongly identified)
##
##
## Multiple arguments have been matched to the same country name:
##   - Eswatini : Eswatini
##   - Iceland : Iceland
##   - IST : Iceland
##   - Swaziland : Eswatini
##   - Turkey : Turkey
##   - Türkiye : Turkey
##
## No close match found for the following countries, kept the original name:
## (set - poor_matches - to TRUE if you want the closest match to be returned or set - na_fill - to TRUE)
## - Authors affiliated with an institute or an international laboratory covered by a cooperation agreement

#For each author, count whether they are only Indian, Only Swiss or both
byauth = dat2 %>%
  group_by(pubNum,author) %>%
  summarise(Swiss = any(country == 'Switzerland'), India = any(country == 'India'),.groups = 'drop',na.rm = TRUE)
  mutate(auth_cat = case_when(Swiss & India ~ 'Both',
    Swiss ~ 'Swiss',
    India ~ 'India', .default = 'None'))

authordat.s = dat2 %>%
  left_join(byauth %>% select(-Swiss,-India), by = join_by(pubNum,author)) %>% #select(-addresses) %>%
  nest(author_details = author:auth_cat,.by=pubNum)

rm(allinst,byauth,dat1,dat2)

```

Now we need to proceed to combine the author data between the two datasets. We are choosing to prioritise

data from Scopus over data from Web of Science, as it has a wider coverage, and is easier to obtain. In case information on a particular publication is not available in Scopus, we will get it from Web of Science.

```
#Give priority to Scopus author data over WoS author data
dat1 = allentries %>%
  filter(datasource != 'WoS') %>%
  left_join(authordat.s,by = 'pubNum')
dat2 = allentries %>%
  filter(datasource == 'WoS') %>%
  left_join(authordat.w,by = 'pubNum')
dat = bind_rows(dat1,dat2)

#Check that all rows have author details
dat_filtered <- dat %>%
  filter(map_lgl(author_details, is.null))
if(nrow(dat_filtered) != 0) {
  printo('Warning: some entries have not gotten author details')
  stop()
}
rm(dat_filtered, dat1,dat2)

#Some Early cleaning
# dat = dat %>% filter(str_detect(`Document Type`,`^(Article|Review)`))

#add in some additional details -
dat = dat |> mutate(nCountries = map_int(author_details, \(authors) length(unique(authors$country))),
  nAuthors = map_int(author_details, \(authors) length(unique(authors$author))),
  nInst = map_int(author_details, \(authors) length(unique(authors$institution))),
  nSwissInst = map_int(author_details, \(authors) authors %>% filter(country == 'Switzerland') %>% nrow()),
  nIndInst = map_int(author_details, \(authors) authors %>% filter(country == 'India') %>% nrow()),
  nSwissAuth = map_int(author_details, \(authors) authors %>% filter(country == 'Switzerland') %>% nrow()),
  nIndAuth = map_int(author_details, \(authors) authors %>% filter(country == 'India') %>% nrow()),
  nBothAuth = map_int(author_details, \(authors) authors %>% filter(auth_cat == 'Both') %>% nrow()))

#For each author, count whether they are only Indian, Only Swiss or both

if(any(dat$nCountries<1) | any(is.na(dat$nCountries))) {
  printo('Warning some entries are not having data parsed down to country level.')
}

#Reduce duplication in columns
dat1 = dat %>%
  filter(datasource == 'WoS') %>%
  rename(`author keywords` = `author keywords.w`, `correspondence address` = `correspondence address.w`)
  select(-`author keywords.s`, -`correspondence address.s`)

saveRDS(dat,file = '~/Data/ISRSD/matched_with_authors.RData')
```